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Roll No.

51477

**B. Pharma. 5th Semester
(Regular/Re-appear)
Examination – December, 2024**

INDUSTRIAL PHARMACY-I

Paper : BP502T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is *compulsory*, each carry 2 marks.

Attempt any *seven* questions out of *nine* from Section-B, each carry 5 marks and *two* questions out of *three* from Section-C, each carry 10 marks.

SECTION – A

Note : Attempt *all* questions.

$2 \times 10 = 20$

1. (a) What are the goal and objectives of Preformulation studies ?

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(b) Define following terms

1. Partition Coefficient.

2. Racemization.

(c) Define quality control test.

(d) Write the manufacturing defects of hard gelatin capsules.

(e) Write a note on type, advantage and limitation of parenteral products.

(f) Discuss equipment for manufacturing of pellets.

(g) Define vanishing cream.

(h) Write the stability studies of pharmaceutical Aerosols.

(i) Differentiate between syrups and elixirs.

(j) Define emulsion and its types.

SECTION – B

Note : Attempt any *seven* questions.

$5 \times 7 = 35$

2. Write a note on BCS classification of drugs and its significance.
- ✓ 3. Discuss the factor affecting stability of suspension.
- ✓ 4. Explain formulation of lyophilized products.
5. Define tablets and discuss about formulation methods and processing problems.
6. Discuss about preparations and formulation defects of Lipsticks.
7. Discuss about manufacturing defect of hard gelatin capsule.
8. Explained tablet coating in detail.
9. Write a note on Emulsion.

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10. Discuss about chemical properties of drug substances.

SECTION – C

Note : Attempt any *two* questions.

2 × 10 = 20

11. Discuss about method of preparations and evaluation of Ophthalmic preparations.

12. Explained in detail about parenteral products.

13. Discuss packaging material science in detailed.

Indira Gandhi University, Meerpur, Rewari

Department of Pharmaceutical Sciences

B.Pharm 5 Semester, 1st Sessional Examination, 2024

Paper Code- BP502T Industrial Pharmacy

Duration: 1 hr 30 minutes

Total marks: 30

Question no.1 is compulsory

- a. Define Preformulation studies
- b. Define polymorphism.
- c. objective of preformulation studies (any 3)
- d. Define emulsion with example
- e. Give two evaluation parameter of liquid orals ✓

2x5=10 marks

2 Attempt any one questions from the rest .

- a. Quality controls test for tablet briefly
- b. Describe all granulation method use for manufacturing of tablet ✓

Wet
Dry
- Power

10x1=10 marks

3 Attempt any two questions from the rest 5x2=10 marks

- a. BCS classification of drug and its significance
- b. Difference between flocculated and deflocculated suspension?
- c. Describe processing problems in tablet manufacturing ✓

Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences

B.Pharma 5 Semester, 2ndnd Sessional Examination, 2024

Paper Code: BP502T Industrial Pharmacy

Duration: 1 hr 30 minutes

Total marks: 30

Question no.1 is compulsory

a. Difference between hard gelatin and soft gelatin

2x5=10marks

b Define propellant with example.

c. Method of preparation of shampoo?

d Two quality control test for capsule?

e. Name component of aerosol (atleast 4)

2 Attempt any one questions from the rest

10x1=10 marks

a. Explain formulation and preparation of toothpaste ,cold cream and lipsticks?

b. Explain briefly about aerosol?

3 Attempt any two questions from the rest

5x2=10 marks

a. Quality control test of hard gelatin capsule ?

b. Explain pellets (pelletization process and formulation requirements)

c. Factors influencing choice of containers and its quality control test?

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**B. Pharma. 5th Semester
(Regular/Re-appear)
Examination – December, 2024**

PHARMACEUTICAL JURISPRUDENCE

Paper : BP505T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is compulsory; each carry 2 marks.

Attempt any seven questions out of nine from Section-B, each carry 5 marks and two questions out of three from Section-C, each carry 10 marks.

SECTION – A

Note : Attempt all questions.

$2 \times 10 = 20$

1. (a) What are the objectives of Drug and Cosmetics Act 1940 ?

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- (b) What is repacking license ?
- (c) What is Schedule P ?
- (d) Highlight the importance of Schedule X in brief.
- (e) What do you mean by First Register ?
- (f) Which types of Ayurvedic preparations containing alcohol ?
- (g) What do you mean by Ceiling Price ?
- (h) Write the formula for the calculation of retail sale price of formulation.
- (i) Write a short note on Design patents.
- (j) Enlist different types of trademarks.

SECTION – B

Note : Attempt any *seven* questions.

$5 \times 7 = 35$

- 2. Which types of drugs are prohibited to manufacture for sale under section 18 of the Drugs and Cosmetics Act.
- 3. Write a note on Offences and Penalties relating to import of drugs.

- ✓ 4. Enumerate conditions of license for manufacture of C₁ and X drugs.
- (Q/R) ✓ 5. What is CDSCO ? Describe the key functions of Drug Inspector.
- ✓ 6. Discuss the significance of Drugs and Magic Remedies (Objectionable Advertisement) Act.
- ✓ 7. Enlist various conditions required for whole sale License.
8. What is Rectified Spirit ? Describe the requirements of a Bonded Laboratory.
- ✓ 9. Write the composition of Institutional Animal Ethics Committee (IAEC).
10. Describe the Pharmacist's Oath.

SECTION - C

Note : Attempt any *two* questions.

2 × 10 = 20

- ✓ 11. Enlist the functions of Pharmacy Council of India.
Discuss the main features of Education Regulations-91.

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12. Describe the purpose of the National List of Essential Medicines (NLEM). How many medicines are listed in current NLEM 2022 ?
13. Write a note on Pharmaceutical Code of Ethics. What are the provisions of Medical Termination of Pregnancy Act ?
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Duration- 1h 30 min

Max. Marks- 30

Question 1 is compulsory. Each part carries 2 marks.

Q1. Write a note on the following-

- a. Cannabis
- b. Narcotic and Psychotropic Substance Drug Consultative Committee
- c. Magic remedy
- d. CPCSEA
- e. Ceiling Price

Attempt any two out of three. Each question carries 5 marks.

Q2. Write the main objectives of Drugs and Magic Remedies Act, 1954.

Q3. Write a short note on Experimentation on animals.

Q4. Give an account of Pharmaceutical Legislation in India.

Attempt any one out of two. Each question carries 10 marks.

Q5. Explain in brief the legal procedure for cultivation of poppy plant (opium)

Q6. What are the requirements for places to be approved for termination of Pregnancy?

Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari

B. Pharmacy 3rd Semester 1st Sessional Examination

Pharmaceutical Jurisprudence (Paper code –BP505T)

Date- 13/09/ 2024

Duration- 1h 30 min

Max. Marks- 30

Q1. Question 1 is compulsory. Each part carries 2 marks.

- i. Define Misbranded Drug.
- ii. Registered Medical Practitioner.
- iii. DCC
- iv. Schedule H
- v. Loan Licence & Repacking Licence
- vi. **Attempt any two out of three. Each question carries 5 marks.**

Q2. Write the conditions & regulations for manufacture of drugs for examination, test and analysis.

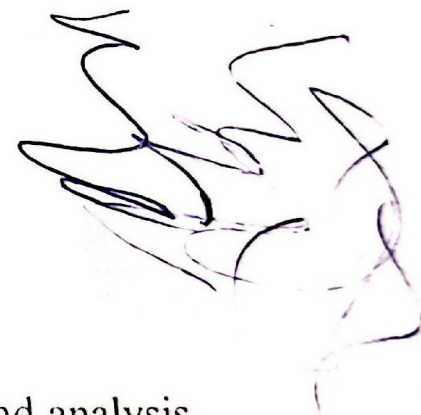
Q3. Mention the classes of drug prohibited to be sold, distributed or exhibited for sale.

Q4. Write a note on Drug Technical Advisory Board.

Attempt any one out of two. Each question carries 10 marks.

Q5. Describe the condition and functions of Pharmacy Council of India.

Q6. Write the duties, qualification & powers of Drug Inspector.



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**B. Pharma. 5th Semester
(Regular/Re-appear)
Examination – December, 2024.**

PHARMACOGNOSY AND PHOTOCHEMISTRY II

Paper : BP504T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is *compulsory*, each parts of question carries 2 marks. Attempt any *seven* questions from Section-B, each carries 5 marks and any *two* questions from Section-C, each carries 10 marks.

SECTION – A

[10 × 2 = 20

1. Answer the followings in brief :

- (a) Write down the Precursors of Acetate melonate pathway.

- (b) What is the chief chemical constituent of Aloe ?
- (c) Write biological source and uses of Curcumin ?
- (d) Write identification methods of Citral.
- (e) What is electrophoresis ?
- (f) Define glycosides. Give its chemical test.
- (g) Write general chemical test of cardiac glycosides.
- (h) Write uses and source of Guggal.
- (i) Describe utilization of Coriander.
- (j) Write industrial production of Forskolin.

SECTION – B**[7 × 5 = 35**

- 2. Discuss various modern methods of extraction.
- 3. Explain acetate mevalonate pathway with structures.
- 4. Write industrial production and estimation of Sennoside.
- ✓ 5. Give utilization and estimation of Caffeine.

6. How will you isolate and identify Atropine ?
- ✓ 7. Write biological source and chemical constituents of any *two* of the followings :
- (i) Gentian
 - (ii) Asafoetida
 - (iii) Clove
- ✓ 8. Write industrial production, estimation and utilization of vincristine and vinblastine.
- ✓ 9. Give source, chemistry and commercial applications of Opium and Ginger.
- ✓ 10. Describe amino acid pathway.

SECTION – C

[2 × 10 = 20

11. Explain the Modern methods of extractions in detail.
12. Give applications chromatographic techniques in the isolation, purification and identification of crude drugs.

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13. Write biological source, chemical constituents and uses of any *four* of the followings :

- (i) Fennel
 - (ii) Almond
 - (iii) Catechu
 - (iv) Digitalis
 - (v) Vinca
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Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari

B. Pharmacy 5th Semester 1st Sessional Examination

PHARMACOGNOSY AND PHYTOCHEMISTRY-I (Paper code -BP 504T)

Date- September 12, 2024

Duration- 1h 30 min

Max. Marks- 30

✓ Q1 is compulsory. Each part carries 2 marks.

Explain in detail

- i. Define secondary metabolites with examples
- ii. Electrophoresis
- iii. Thin layer chromatography
- iv. NMR spectroscopy
- v. Amino acid pathway ✓

cell Attempt any two out of three. Each question carries 5 marks.

Q2. Explain High performance liquid chromatography in detail.

Q3. Explain the role of radioactive isotopes in investigation of biogenetic pathway.

Q4. Explain UV-visible and IR spectroscopy in detail.

Attempt any one out of two. Each question carries 10 marks.

Q5. Explain Shikmic acid pathway and acetate mevalonate pathway with structures.

Q6. What are the difference between traditional and modern method of extraction. Explain modern methods of extraction in detail.

Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari

B. Pharmacy 5th Semester 2nd Sessional Examination

PHARMACOGNOSY AND PHYTOCHEMISTRY-II (Paper code -BP 504T)

Date- November 03, 2024

Duration- 1h 30 min

Max. Marks- 30

Q1 is compulsory. Each part carries 2 marks.

Explain in detail

- i. Uses of Rauwolfia
- ii. Caffeine
- iii. Examples of Terpenoids
- iv. Define Alkaloids and its functions
- v. Chemical constituents of Mentha

Attempt any two out of four. Each question carries 5 marks.

Q2. Explain isolation, identification, and analysis of Citral.

Q3. Write biological source, chemical constituents and uses of Cinnamon.

Q4. Explain isolation, identification, and analysis of Curcumin.

Q5. Write biological source, chemical constituents and uses of Ginger.

Attempt any one out of two. Each question carries 10 marks.

Q6. Explain isolation, identification, and analysis of Menthol and Atropine.

Q7. Write biological source, chemical constituents and uses of Vinca and Coriander

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**B. Pharma. 5th Semester
(Regular/Re-appear)
Examination – December, 2024**

PHARMACOLOGY - II

Paper : BP503T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is compulsory, each carry 2 marks. Attempt any seven questions out of nine from Section-B, each carry 5 marks and two questions out of three from Section-C, each carry 10 marks.

SECTION – A

Note : Attempt all questions.

$2 \times 10 = 20$

1. (a) Define Effective Refractory period.

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- (b) Describe the mechanism of action of Nicorandil as anti-anginal drug.
- (c) What are Haematinics ? Give examples.
- (d) Name any two P2Y₁₂ receptor blockers.
- (e) Write the mechanism of action of Renzapride.
- (f) Name the comorbid conditions aggravated by NSAIDs.
- (g) Name two hormones secreted from acidophil cells of the anterior pituitary.
- (h) Write the mechanism of action of Sitagliptin.
- (i) What are anabolic steroids ? Give examples.
- (j) Enlist the clinical uses of Mifepristone.

SECTION – B

Note : Attempt any *seven* questions.

5 × 7 = 35

- 2. Justify the therapeutic role of Phosphodiesterase inhibitors and vasodilators in Congestive heart failure.
- 3. What are low molecular weight heparins ? Describe the mechanism of action of Dipyridamole.

4. Discuss the synthesis, storage and destruction of 5-HT. Enlist different types of 5-HT receptors.
5. Discuss the synthesis, storage and destruction of Thyroid hormone.
6. Discuss the chemical and hormonal regulation of insulin secretion.
7. Classify Corticosteroid. Write the mechanism of action of Sulfonylureas.
8. Illustrate the physiological actions of Testosterone.
9. Discuss bioassay of Oxytocin. 10:46
10. Write a note on Hormonal contraceptives.

SECTION - C

Note : Attempt any *two* questions.

2 × 10 = 20

11. Classify antihypertensive drugs. Explain the mechanism of action of Methyldopa and Nifedipine as antihypertensive drugs.

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12. Classify anticoagulants and Write a note on Erythropoietin.
 13. Describe the distinctive features of different types of Histamine receptors in detail.
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Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B. Pharm. 5th Semester, 1st Sessional Examination, September, 2024
Paper Code – BP503.T; Pharmacology- II (Theory)

Duration: 1 hr 30 minutes

Total marks: 30

1. Question no. 1 is compulsory.

(5×2=10 marks)

Write a short note on followings:-

- Electrophysiology of heart
- Anterior Pituitary hormones and their action
- Function of Vitamin-D
- Function of Oxytocin
- Applications of bioassay

2. Attempt any two questions from the rest.

(2×5=10 marks)

- Discuss about synthesis and normal physiology of Androgen hormone and related drugs?
- Describe about pharmacology of Thyroid hormones?
- Explain different types of bioassay?

3. Attempt anyone questions from the rest.

(1×10=10 marks)

- Describe in details about Hypertension and Anti-hypertensive drugs?
- Describe in details about pharmacology of Insulin and Oral Hypoglycemic agents?

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B. Pharmacy 5th Semester 2nd Sessional Exam Nov. 2024
Pharmacology II Subject code: BP503T

Time: 1.5 Hrs

Question 1 is compulsory. Each question carry 2 marks.

Max. Marks: 30 *fenar*

*fenamic
acid*

*Acetic acid derivative
- Prostaglandin*

Q1. Write a note on the following-

- a. Plasma volume expander
- b. Anticoagulant
- c. Antidiuretics
- d. Fibrinolytics
- e. Classification of NSAID.

Attempt any two out of three. Each question carries 5 marks.

Q2. Classify diuretics with example.

Q3. Describe the pharmacology of statins and calcium channel blockers.

Q4. Write a note on Histamine pharmacology..

Attempt any one out of two. Each question carries 10 marks.

Q5. Classify anti arrhythmic drugs with example. Describe pharmacology of quindine sulphate.

Q6. Detail note on Prostaglandins, Thromboxanes and Leukotrienes.